Ali Shibli

PhD Researcher | Computer Vision, Representation Learning, Foundation Models

Stockholm, Sweden shibli@kth.se +46 724 416 295

GitHub Google Scholar Portfolio LinkedIn

Research focus. Visual representation learning and foundation models for Earth observation; self-supervised learning, geographic priors, generative modeling, and robust adaptation across geographic and sensor shifts.

Expected PhD graduation: May 2028.

Education

PhD in Geoinformatics *May 2024–expected May 2028*
KTH Royal Institute of Technology, Stockholm, Sweden. Advisors: Yifang Ban and Andrea Nascetti.
Research: foundation models and representation learning for Earth observation.

M.Sc. in Machine Learning, KTH Royal Institute of Technology *2020–2022*
B.Eng. in Computer and Communications Engineering, American University of Beirut *2015–2020*
B.Sc. in Mathematics, American University of Beirut *2016–2020*

Research and Industry Experience

PhD Researcher | KTH Royal Institute of Technology, Stockholm *May 2024–present*

- Developed **GeoDistill**, a self-distillation continual learning method that uses georeferenced neighboring imagery to transfer wider-context teacher features into DINOv3 for geographic data and improves downstream tasks.
- Developed **Noise2Map**, a diffusion-inspired framework for segmentation and change detection with single-pass prediction; published in *IEEE Transactions on Geoscience and Remote Sensing*; released implementation and weights.
- Studied parameter-efficient adaptation of TerraMind, DINOv3, and Prithvi-EO-2.0 for wildfire mapping across 3,820 events; LoRA improved generalization under geographic and temporal shift while updating less than 1% of encoder parameters. Presented at IGARSS and the ECCV TerraBytes II workshop in 2026.
- Contributed to **PANGAEA**, a published global benchmark for evaluating geospatial foundation models across tasks, image resolutions, and sensing modalities.

Data Scientist I–II | Ericsson Global AI Accelerator, Stockholm *Jan 2022–Apr 2024*

- Developed computer vision and time-series models for telecom network optimization; improved forecasting accuracy by 5% through architecture search and feature engineering.
- Built RLDevOps, a Google Cloud-based MLOps framework for reinforcement learning workflows. Named inventor on seven AI and telecom patent applications filed during the role.

ML Research Intern | Ericsson, Stockholm *Jun–Dec 2021*
Developed NLP and collaborative-filtering recommender systems for internal knowledge management.

ML Research Engineer | KTH Royal Institute of Technology *Jan–May 2021*
Built graph neural networks for visual recognition and an image-caption collection tool with 12,000+ PyPI downloads.

Selected Publications

- **A. Shibli**, A. Nascetti, Y. Ban. "Noise2Map: End-to-End Diffusion Model for Semantic Segmentation and Change Detection." *IEEE Transactions on Geoscience and Remote Sensing*, 2026.
- V. Marsocci, Y. Jia, et al. (including **A. Shibli**). "PANGAEA: Assessing Geospatial Foundation Models Capabilities through a Global and Inclusive Benchmark." *IEEE Geoscience and Remote Sensing Magazine*, 2026.
- **A. Shibli**, A. Nascetti, Y. Ban. "Low-Rank Adaptation of Geospatial Foundation Models for Wildfire Mapping Using Sentinel-2 Data." *IGARSS 2026*; poster at *ECCV 2026 TerraBytes II Workshop*.
- **A. Shibli**, A. Nascetti, Y. Ban. "Very High- to High-Resolution Imagery Transferability for Building Damage Detection Using Generative AI." *JURSE 2025*.

Selected Awards and Research Support

- Top-10 finalist, IGARSS Three Minute Thesis (3MT) competition, 2026.
- IGARSS 2026 travel grant for presenting research in Washington DC; Kartografiska Sällskapet travel scholarship, 2026.
- First place, CVPR Embed2Scale Challenge, 2025; third-best student paper, JURSE, 2025.
- First place, Final-Year Engineering Project Competition, American University of Beirut (AUB).

Technical Skills

Languages and frameworks: Python, C++, Java, SQL; PyTorch, TensorFlow.
Tools and methods: Git, Linux, SLURM, Google Cloud; computer vision, self-supervised learning, diffusion models, multimodal learning, evaluation under domain shift.